



Deep Learning

Lecture 5.2: Brief Introduction of Other Networks



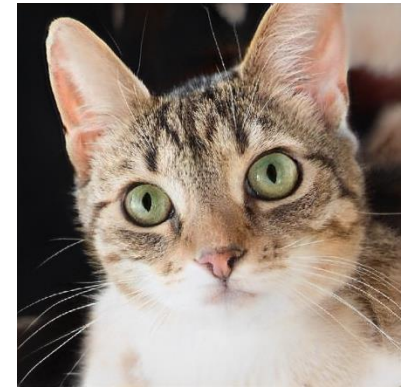
Generative Models

Model the data distribution or conditional data distribution:

$$P(x) \text{ or } P(x | y)$$

Why do we care?

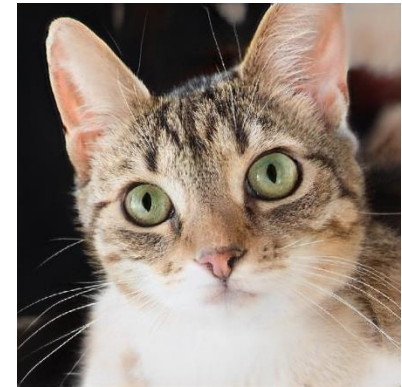
- Having a density estimate lets you check the probability of a new sample
- Having a density estimate lets you sample from the distribution





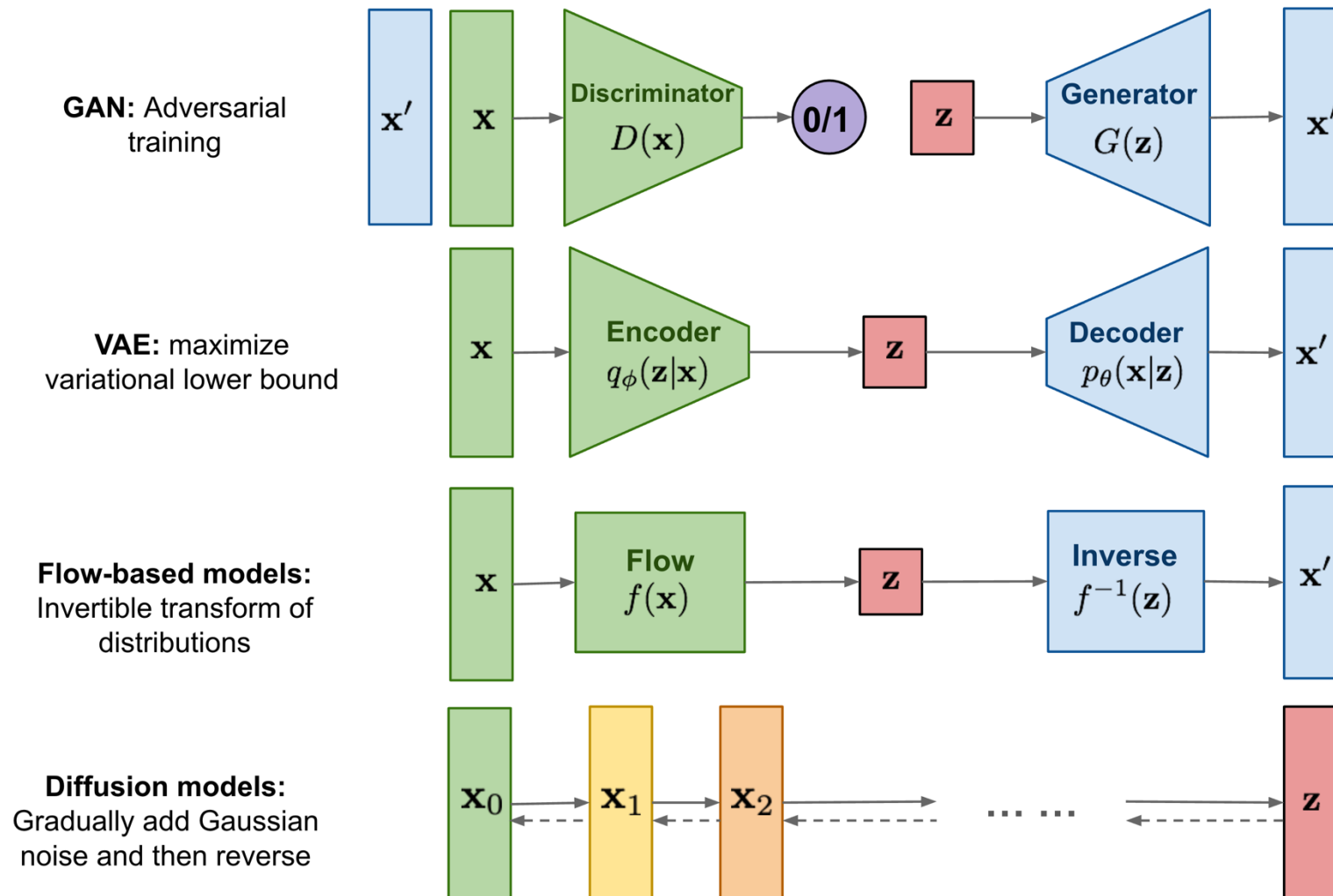
Traditional methods for density estimation in ML fall short in high dimensional, structured domains.

- For example, the space of 256×256 images is HUGE and the correlations between pixels are very complex.
- Fit it with a Gaussian? A mixture of Gaussians? A kernel density estimate?



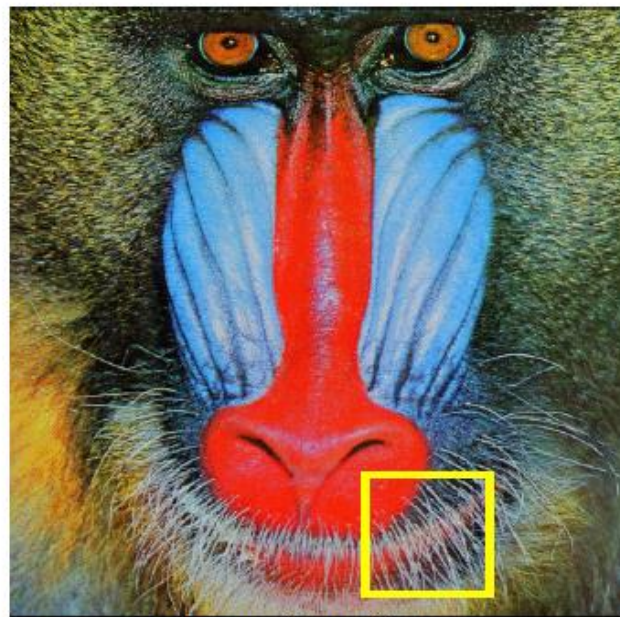
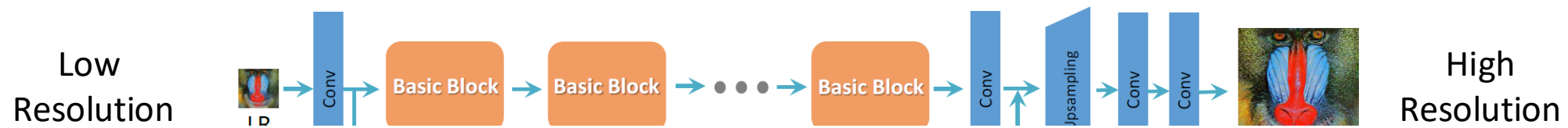


A number of model architectures / training regimes have been introduced





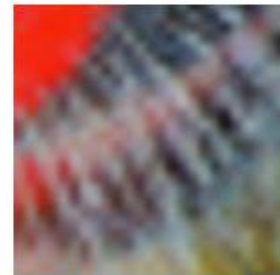
Super-resolution: Some result comparisons



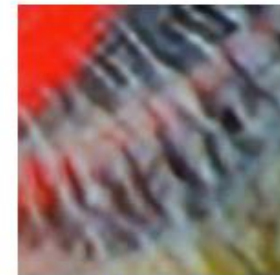
baboon from Set14
(PSNR / Perceptual Index)



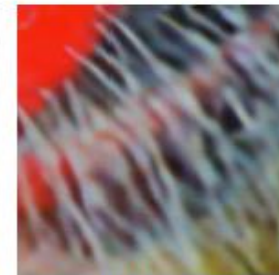
HR
(∞ / 3.59)



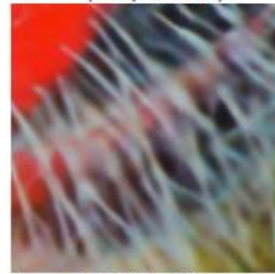
Bicubic
(22.44 / 6.70)



SRCNN
(22.73 / 5.73)



EDSR
(23.04 / 4.89)



RCAN
(23.12 / 4.20)



EnhanceNet
(20.87 / 2.68)



SRGAN
(21.15 / 2.62)



ESRGAN(ours)
(20.35 / 1.98)



GAN-Based Image Inpainting



Figure 1: Example inpainting results of our method on images of natural scene, face and texture. Missing regions are shown in white. In each pair, the left is input image and right is the direct output of our trained generative neural networks without any post-processing.



GAN-Based Image Deblurring

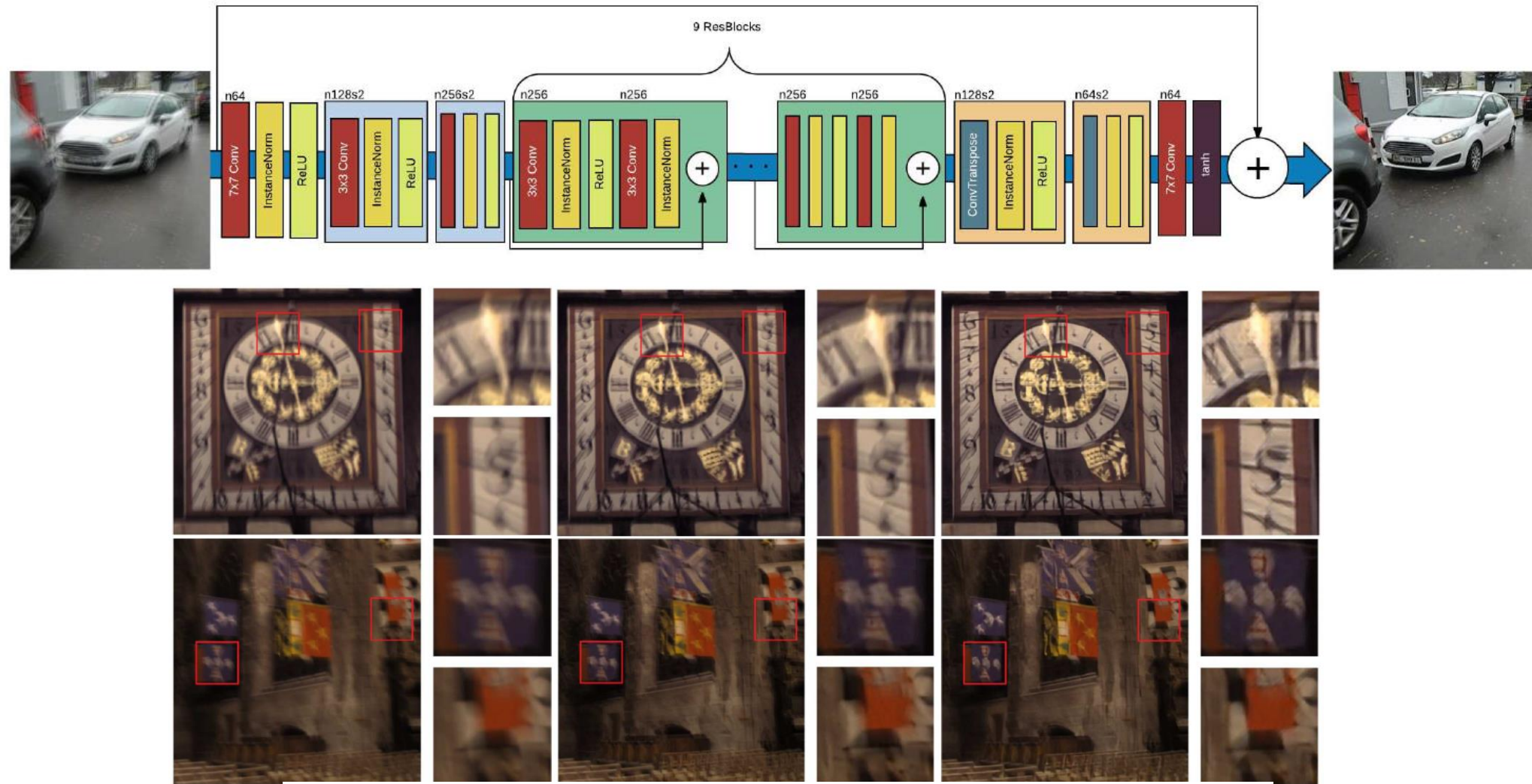
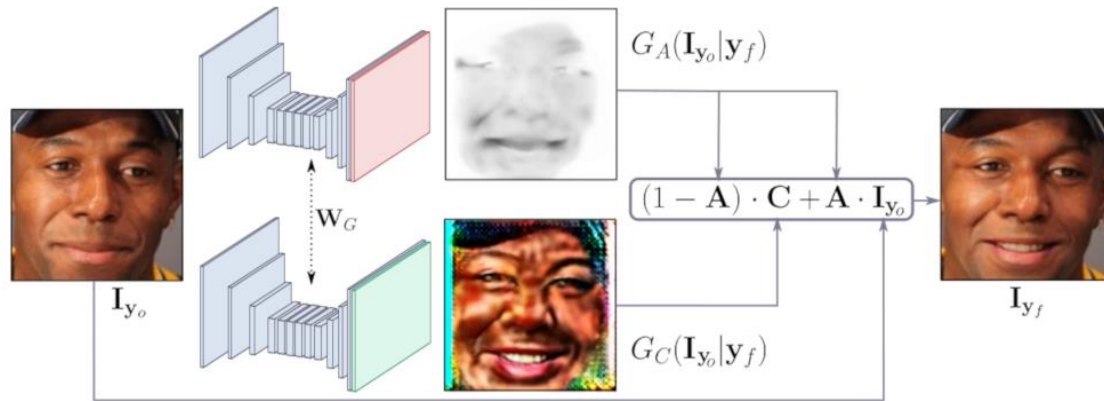


Figure 8: Results on the Kohler dataset. From left to right: blurred photo, Nah *et al.* [25], DeblurGAN.



GANimation: Facial Expression Changes



<https://www.albertpumarola.com/research/GANimation/>



Text-to-Image: Condition on Natural Language Features

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



the flower has petals that are bright pinkish purple with white stigma



this white and yellow flower have thin white petals and a round yellow stamen



Figure 23: Text-to-image synthesis with GANs. Image reproduced from Reed *et al.* (2016b).



Use multiple steps to learn for more details

<https://stablediffusionweb.com/>



What have we done so far

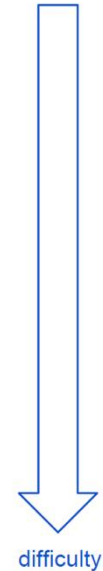
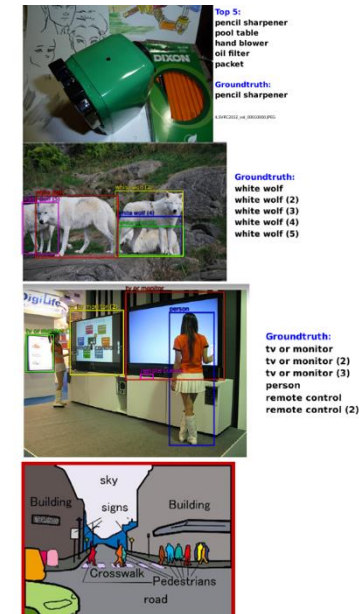
Mostly image tasks:

- classification

- localization

- detection

- segmentation



Discriminative tasks:

Generative tasks:

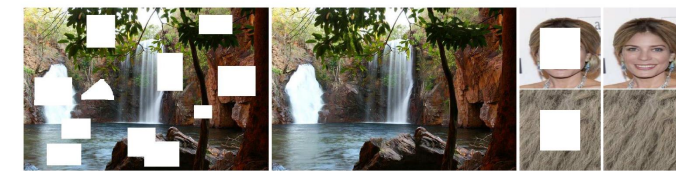
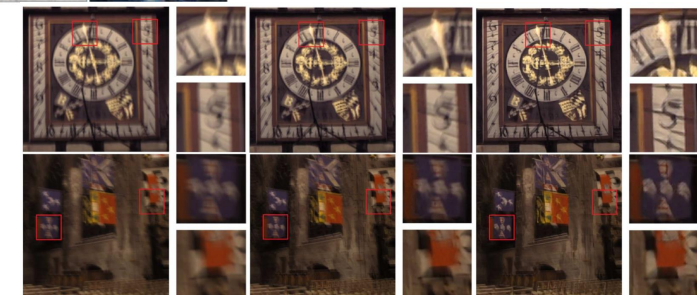


Figure 1: Example inpainting results of our method on images of natural scene, face and texture. Missing regions are shown in white. In each pair, the left is input image and right is the direct output of our trained generative neural networks without any post-processing.





First there are some image-based tasks that might not be that obvious

Game playing:



Earlier: Atari games, later: AlphaStar etc.

Board Games:



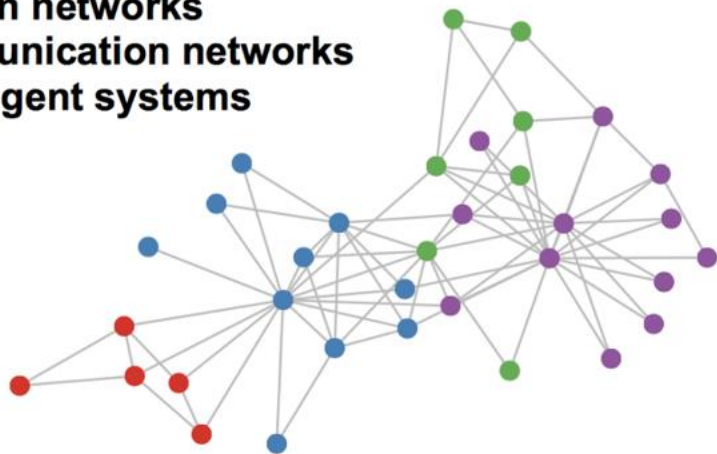
Board games can be treated as binary images



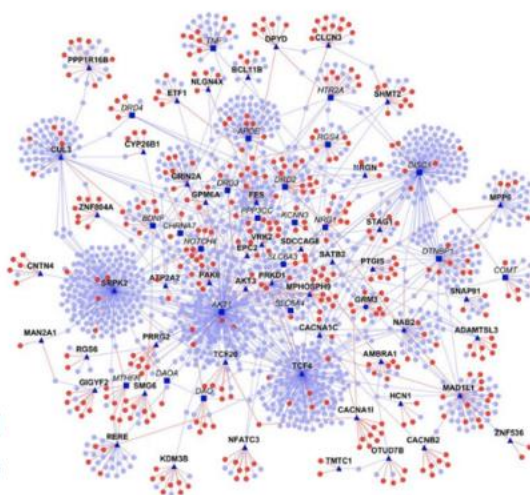
What about non-regular domains? One idea is graphs

Many problems don't lie on regular grids

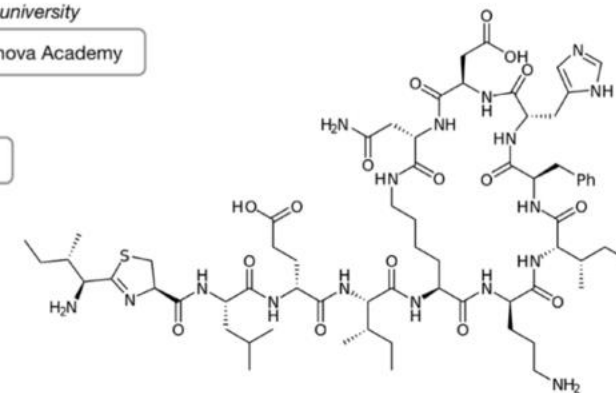
Social networks
Citation networks
Communication networks
Multi-agent systems



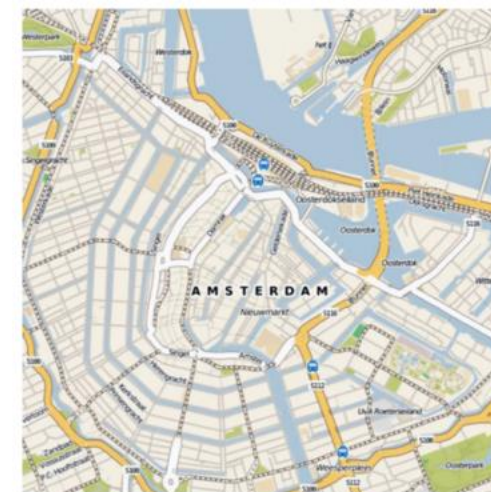
Protein interaction networks



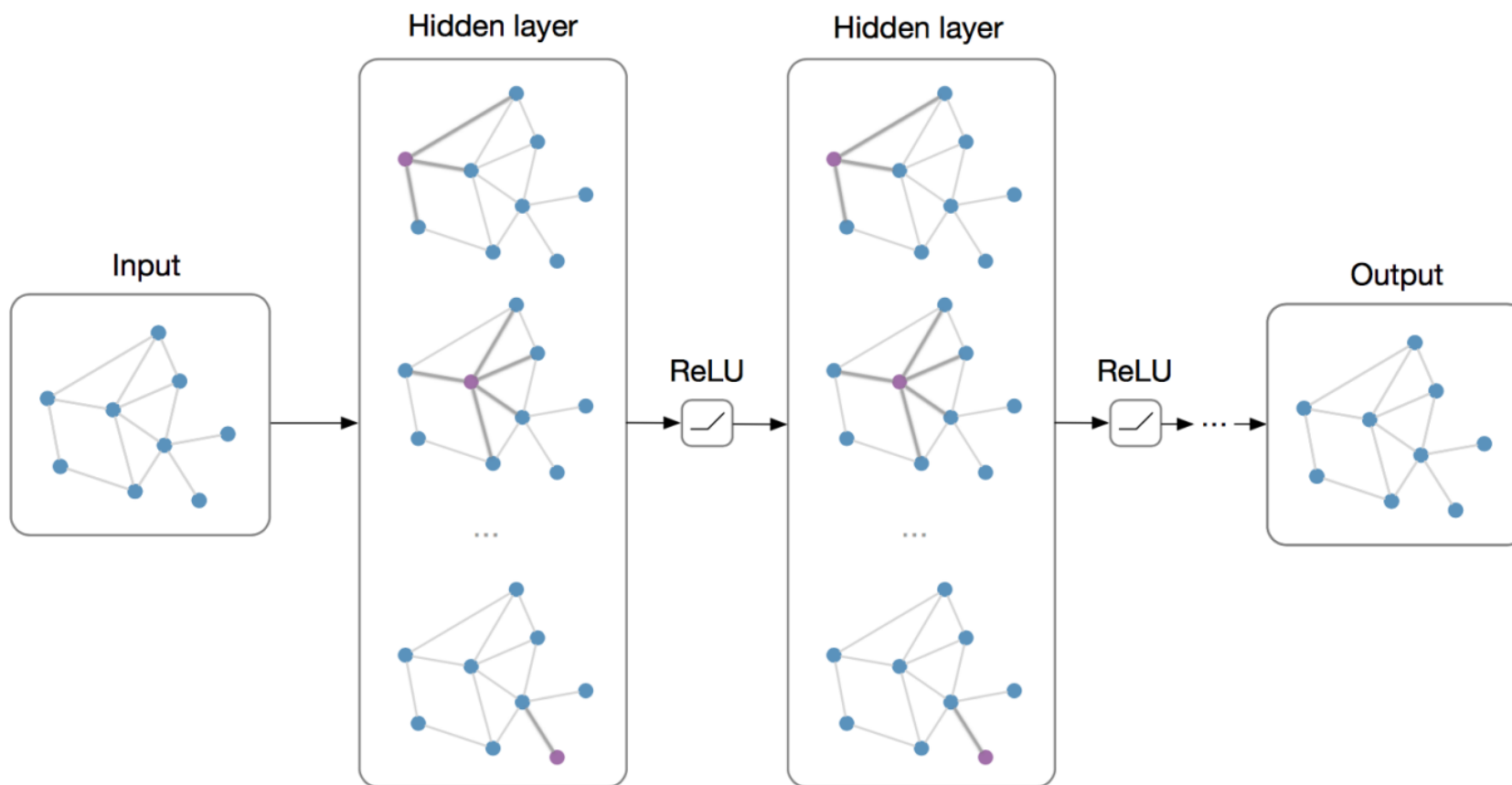
Knowledge graphs



Molecules



Road maps



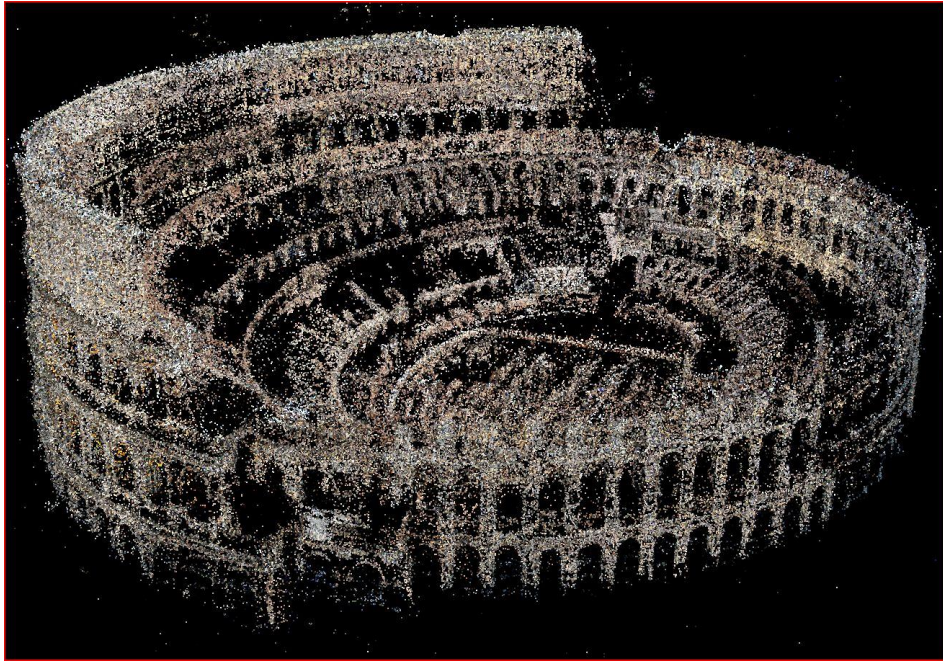
Strategy: Each node (and maybe edge) has some initial feature. These features get updated based on other features and the connectivity in the graph.



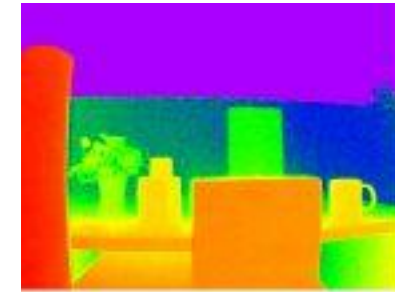
Another idea is Point Clouds

Irregular points in low-dimensional space

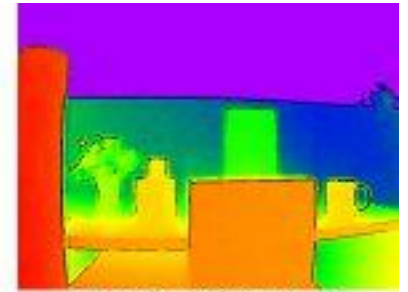
LIDAR Data



RGB-D



Normal depth frame



Depth frame after applying post processing filters

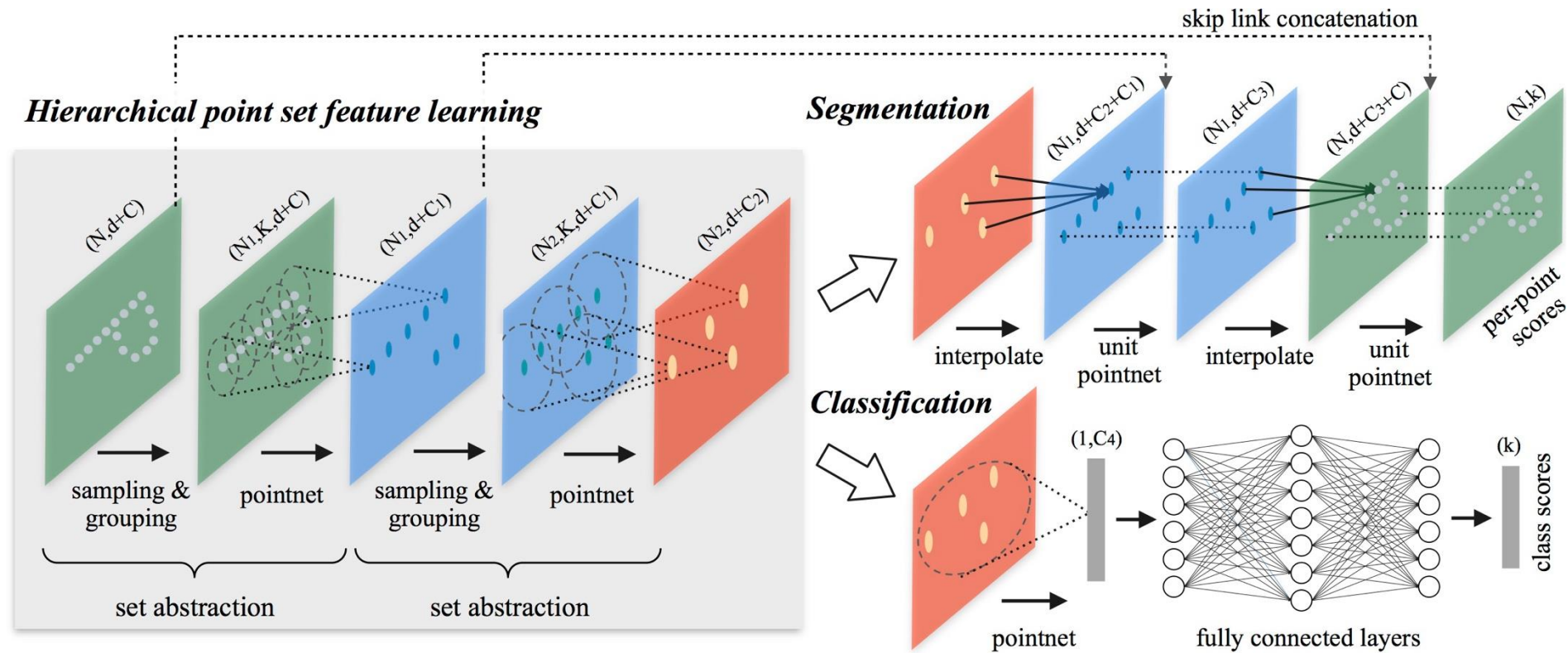
As a general 3D data representation format

Video Demo



Similar Kinds of Deep Network Can be Built for Point Clouds Too

Similar ideas of convolutions/downsampling/upsampling can be defined





We will describe the Transformer model used in GPTs

